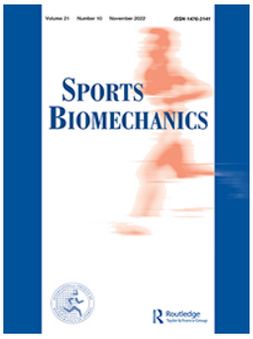

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Joint-level determinants of the dynamic strength index: implications for testing and monitoring

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ABSTRACT

The aims of this study were to determine the associations and differences between the traditional dynamic strength index (DSI_{GRF}, calculated from ground reaction force [GRF] data) and joint-level dynamic strength indices (DSI_{NJM}, calculated from net joint moment [NJM] data). Eight female NCAA Division I lacrosse players performed three maximal effort isometric mid-thigh pulls (IMTP) and countermovement jumps (CMJ). GRF and motion capture data were recorded and used to calculate hip, knee, and ankle NJM. The ratio between peak forces during the IMTP and CMJ tests was used to calculate the DSI_{GRF} and the ratio of peak NJM was used to calculate ankle, knee, and hip DSI_{NJM}. Associations and differences between DSI_{GRF} and DSI_{NJM} were examined with Pearson's correlation coefficients, paired *t*-tests and Hedge's *g* effect sizes. The results showed that only the hip DSI_{NJM} was correlated with DSI_{GRF} (hip: $p = 0.037$; $r = 0.738$). In addition, athletes exhibited significantly greater DSI_{NJM} than DSI_{GRF}. Although these results suggest DSI_{GRF} are associated with hip DSI_{NJM}, the fact that joint-level DSI_{NJM} were generally greater than DSI_{GRF} suggests the ratio between peak NJM during the CMJ and IMTP may not provide intuitive or useful data for strength and conditioning coaches or sport scientists.

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Biomechanics; isometric mid-thigh pull; countermovement jump; dynamic strength deficit

Introduction

Increasing maximal strength is thought to be essential for improving athletic performance in dynamic athletic tasks such as jumping, sprinting, and changing of direction (Haff et al., 1997; Nuzzo et al., 2008; Stone et al., 2003; Suchomel et al., 2016; West et al., 2011). An athlete's maximal strength, or their maximum force producing capacity, is commonly quantified with either dynamic or isometric testing (Comfort et al., 2012; De Witt et al., 2018; Grgic et al., 2020; Haff et al., 1997). Many studies assess an athlete's maximum force producing capacity with the isometric mid-thigh pull (IMTP) because it is safe, simple, and thought to cause less fatigue than other maximal strength tests (e.g., one-repetition maximum) (De Witt et al., 2018; Haff et al., 1997; Kawamori et al., 2006). In addition to such testing, many studies also assess how effectively athletes use their

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maximal strength during dynamic tasks, such as countermovement jumps (CMJ) (Comfort et al., 2018; Secomb et al., 2016; Sheppard et al., 2011; Thomas et al., 2015). Specifically, these studies calculate the ratio in peak ground reaction forces (GRF) during the CMJ and IMTP, which is often referred to as the dynamic strength index (DSI) or dynamic strength deficit (DSD).

Studies suggest that the DSI is a valid and reliable measure that can be used to evaluate an athlete's strength qualities and monitor training-induced changes (Comfort et al., 2018, 2018; McMahon et al., 2017; Sheppard et al., 2011; Thomas et al., 2015). The DSI has also been used to inform current training status and guide future training interventions (Comfort et al., 2018; McMahon et al., 2017). Specifically, researchers and practitioners used the DSI to identify whether an athlete would benefit from maximal strength, speed-strength, or combined (i.e., maximal and speed-strength) training (McMahon et al., 2017; Sheppard et al., 2011). In this context, a DSI of 100% indicates that an athlete can effectively utilise all their maximum strength during the dynamic performance of that task. In addition, DSI values between 60% and 80% are thought to reflect 'balanced' DSI values, where an athlete with a DSI below 60% may not effectively utilise their maximum strength during dynamic performance and may benefit from focusing on producing greater forces during dynamic tasks (Sheppard et al., 2011).

Many earlier researchers calculated DSI values from ground reaction force (GRF) data (Comfort et al., 2018, 2018; Sheppard et al., 2011). While GRF data from tests such as the IMTP provide insight about an athlete's whole-body maximal strength, GRF magnitudes are not always associated with net joint moments (NJM), which implies that evaluating maximal strength during multi-joint isometric exercises through only GRF data offers limited information about joint-specific strength (Hahn, 2011). For example, Hahn showed that people could produce similar GRF magnitudes but with different combinations of NJM contributions from the ankle, knee and hip extensors during an isometric leg press. Similarly, other authors found that during concentric phase of the CMJ some people produce the greatest joint moments at the hip while others produce greater joint moments at the knee (Cleather et al., 2013; McErlain-Naylor et al., 2014; Vanezis & Lees, 2005; Williams et al., 2018). While these findings reflect subject-specific strategies, they may also reflect muscle-specific strength capacities, which may ultimately lead to different joint-specific contributions to GRF. Since GRF data may not fully reflect the functional demands or contributions from specific muscle groups, GRF-based DSI (DSI_{GRF}) values may not fully reflect how effectively an athlete uses the strength capacity of specific muscles. Therefore, determining the relationship between DSI_{GRF} and joint-based DSI (DSI_{NJM}) values would provide specific information about whether the DSI_{GRF} represents a valid assessment of joint-specific strength qualities.

To our knowledge, no previous studies investigated DSI_{NJM}, which means that it is not known if DSI_{GRF} and DSI_{NJM} are correlated, and whether DSI_{GRF} is a valid surrogate of joint-specific strength. Calculating DSI_{NJM} of the major extensor joint of the lower body and determining their correlation with DSI_{GRF} could help expand our understanding about how athletes use their neuromuscular capacities during dynamic tasks (i.e., identify which joints contribute to DSI_{GRF}) and provide useful information for planning training programmes. Therefore, the aims of this study were to determine the associations and differences between DSI_{GRF} and DSI_{NJM} values. We hypothesised that the traditional

DSI_{GRF} values would correlate with at least one DSI_{NJM} and that these DSI values would not differ from each other.

Materials and methods

Subjects

To determine the necessary number of subjects an a priori power analysis was performed. Since the purpose of the study focused on the associations between DSI_{GRF} and DSI_{NJM}, a correlation (i.e., bivariate normal model) was used as the statistical test for the power calculation. Further, it was reasoned that if DSI_{GRF} values were indeed valid and meaningful predictors of DSI_{NJM} values then the correlation coefficients would be between 0.70 and 0.90 (i.e., very large). Therefore, the a priori power calculation for a correlation analysis indicated that to obtain very large correlation coefficient (Pearson's r) of 0.70–0.90 and an effect size (coefficient of determination – ρ^2) of 0.5–0.8 with an alpha level of 0.05 and statistical power of 0.80 a minimum sample size of 6–13 would be required (G*Power 3.1, Universität Düsseldorf, Germany). Ten female NCAA Division I lacrosse players (age: 20 ± 2 years; height: 1.70 ± 0.03 m; body mass: 65.4 ± 5.9 kg) were initially recruited to participate in this study, but data from two of the players had to be discarded due to unanticipated problems. All players were familiar with the IMTP and CMJ procedures through participation in an athlete monitoring programme and previous research studies. Data collection for the current study occurred after the end of the player's off-season training programme. The study was approved by the local University's Institutional Review Board, and all players provided written informed consent before beginning their participation in the current study.

Testing procedures

Each player was outfitted with 28 reflective markers, which were attached to anatomical landmarks on the pelvis, thighs, shanks, and feet of both legs. Specifically, markers were attached to easily identifiable anatomical protuberances, i.e., bi-lateral anterior and posterior superior iliac spines, iliac crests, greater trochanters, femoral epicondyles, fibular heads, tibial tuberosities, malleoli, calcanei, metatarsal heads (1st and 5th) and bases (5th).

After marker attachment, each player performed a standardised dynamic warm-up that consisted of simple callisthenic exercises and several submaximal and maximal CMJ. Players performed three maximal effort CMJ with at least 30 seconds rest between jumps. During each CMJ, players were instructed to place their hands on their hips and jump as high and as fast as possible. Athletes could choose their preferred CMJ depth. For IMTP testing, each athlete performed the IMTP with the bar locked into an IMTP rack (Kairos Strength) and positioned at mid-thigh, i.e., with knee and hip angles of approximately 145 and 135 degrees, respectively. Joint angles were standardised with a goniometer. Athletes then performed two separate submaximal IMTP trials at progressively higher effort levels, which were followed by three maximal effort trials. Neither hands nor wrists were strapped to the bar. During each maximal effort IMTP trial, athletes were instructed to pull on the bar as hard and as fast as possible, and to pull for at least three seconds.

Athletes received strong verbal encouragement during each maximal effort IMTP trial. A rest period of at least one minute was provided between each submaximal and maximal effort.

Data collection procedures

Kinematic data were collected with 14 motion capture cameras (T Series; Vicon, Los Angeles, CA) at 100 Hz. Kinetic data during the IMTP were recorded with two portable force plates (Kistler, Winterthur, Switzerland) at 1000 Hz. Kinetic data during the CMJ were recorded with two in-ground force plates (AMTI, Watertown, MA, USA) at 1000 Hz. Kinematic and kinetic data were synchronously collected with the motion capture system's software (Nexus 1.8.2; Vicon, Los Angeles, CA, USA) and exported as .c3d files.

The .c3d files were imported and processed with Visual3D software (C-Motion, Inc., Rockville, MD, USA). Kinematic and kinetic data were filtered with a fourth-order low-pass Butterworth filter at a cut-off frequency of 18 Hz for the CMJ and 8 Hz for the IMTP, which were determined based on a residual analysis. The filtered data were used as inputs to a custom biomechanical model that consisted of pelvis, thigh, shank, and foot segments. A static trial was used to define segment coordinate systems based on anatomical markers of the proximal and distal ends of each respective segment. Inverse kinematic procedures were used to minimise the effect of soft tissue and measurement errors when estimating the position and orientation of the segments during the IMTP and CMJ (Lu & O'Connor, 1999). Internal NJM at ankle, knee, and hip joints were calculated and processed such that positive values represent hip extension, knee extension and ankle plantarflexion moments.

Peak GRF and NJM were extracted from the pull phase of the IMTP trials and from the concentric phase of the CMJ, which was defined as the point at which the hip, knee, and ankle angles had reached maximum flexion and when all joints changed from negative to positive joint power (i.e., transition from eccentric to concentric contraction). Peak data were averaged across the left and right leg as well as across all three trials (of either the IMTP or CMJ trials). The peak GRF and NJM were normalised to body-mass (i.e., N/kg and N·m/kg, respectively). DSI_{GRF} (Equation (1)) and DSI_{NJM} (Equation (2)) DSIs were calculated with the following equations and used for statistical analyses:

$$DSI_{GRF} = \frac{CMJ\ Peak\ GRF}{IMTP\ Peak\ GRF} \tag{1}$$

$$DSI_{NJM} = \frac{CMJ\ Peak\ NJM_{Hip,Knee,Ankle}}{IMTP\ Peak\ NJM_{Hip,Knee,Ankle}} \tag{2}$$

Statistical analyses

All data are presented as mean ± standard deviation (SD). Normality was assessed with the Shapiro—Wilk test and Q-Q plots. The reliability of DSI_{GRF} and DSI_{NJM} was evaluated via the intraclass correlation coefficient (ICC (3, k)) and interpreted as poor (ICC <0.5), moderate (ICC = 0.5–0.75), good (ICC = 0.75–0.9), and excellent (ICC >0.9)

Table 1. Correlation coefficients (Pearson's *r*) with 95% confidence intervals (upper bound, lower bound) between ground reaction force (GRF) and net joint moment (NJM) dynamic strength indices (DSI).

	DSI_{NJM}		
	Ankle	Knee	Hip
DSI_{GRF}	−0.349 (−0.937, 0.915)	0.139 (−0.821, 0.865)	0.738 (0.147, 0.976) *

Note: * $p < 0.05$.

(Koo & Li, 2016). The coefficient of variations (CV%) of the data were also calculated and interpreted as acceptable if <10% (Bishop et al., 2021; Cormack et al., 2008; Currell & Jeukendrup, 2008). Associations between DSI_{GRF} and DSI_{NJM} values were investigated with Pearson correlation coefficients. Bootstrapping (n = 100) was used to calculate 95% confidence intervals for each set of correlation results. The strengths of correlation coefficients were interpreted as 0.1–0.3 = small; 0.3–0.5 = moderate; 0.5–0.7 = large; 0.7–0.9 = very large; 0.9–1 = perfect (Hopkins, 2006). DSI_{GRF} and DSI_{NJM} values were compared with paired *t*-tests. Further, the mean difference for all comparisons along with the 95% confidence intervals of the respective differences were calculated. Lastly, effect sizes (Hedge's *g*) for pairwise comparisons were calculated and interpreted as either small (0.20–0.49), moderate (0.50–0.79), or large (≥0.80) (Hedges, 1985; Lakens, 2013). The level of statistical significance was set to an α-level of 0.05. All statistical analyses were performed in SPSS 26.0 (IBM Corp, Armonk, NY, USA).

Results

The intraclass correlation coefficients (ICC [95% confidence interval]) of the DSI_{NJM} and DSI_{GRF} ranged from 0.960 to 0.993 (DSI_{GRF}: 0.960 [0.850–0.992]; DSI_{Ankle}: 0.993 [0.974–0.999]; DSI_{Knee}: 0.993 [0.974–0.999]; DSI_{Hip}: 0.969 [0.884–0.994]). The coefficients of variation (CV% [95% confidence intervals]) of the data ranged from 3.2% to 6.8% (DSI_{GRF}: 3.2% [1.2–5.1]; DSI_{Ankle}: 3.5% [1.3–5.8]; DSI_{Knee}: 4.4% [2.0–6.8]; DSI_{Hip}: 6.8% [3.6 – 10]).

DSI_{GRF} and DSI_{NJM} – correlation

Only the hip DSI_{NJM} exhibited a significant ($p = 0.037$) and very large positive correlation with DSI_{GRF} (Table 1; Figure 1).

DSI_{GRF} and DSI_{NJM} – comparison

All average DSI_{NJM} were significantly greater than the DSI_{GRF} and except for one joint/subject all DSI_{NJM} values exceeded all DSI_{GRF} values (Figure 2). Specifically, pair-wise comparisons indicated that ankle (difference = 34.3 [8.8, 59.7]; $p = 0.016$; Hedges $g = 1.57$), knee (difference = 62.1 [14.9, 109.3]; $p = 0.017$; Hedges $g = 1.35$), and hip (difference = 111.5 [70.9, 152.1]; $p = 0.001$; Hedges $g = 2.49$) DSI_{NJM} were all statistically and practically greater than DSI_{GRF}. Analysis of DSI_{NJM} and DSI_{GRF} values for individual subjects indicated the presence of intra-individual variability in DSI_{NJM} ratios (Figure 3).

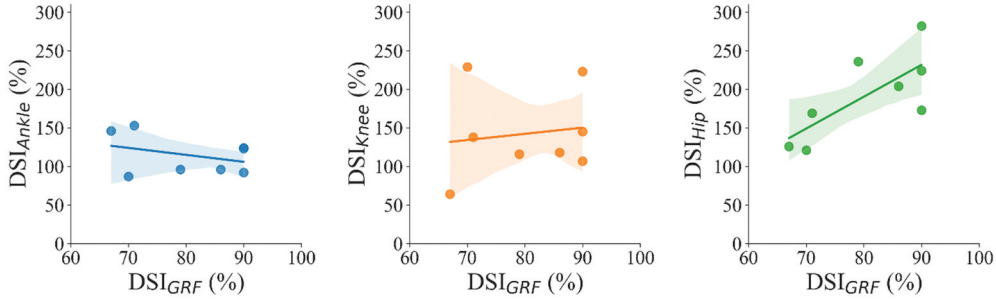

Figure 1. Relationship (Pearson's *r* with 95% confidence interval) between ground reaction force based (DSI_{GRF} [%]) and ankle, knee, hip net joint moment based dynamic strength indices (DSI_{NJM} [%]).

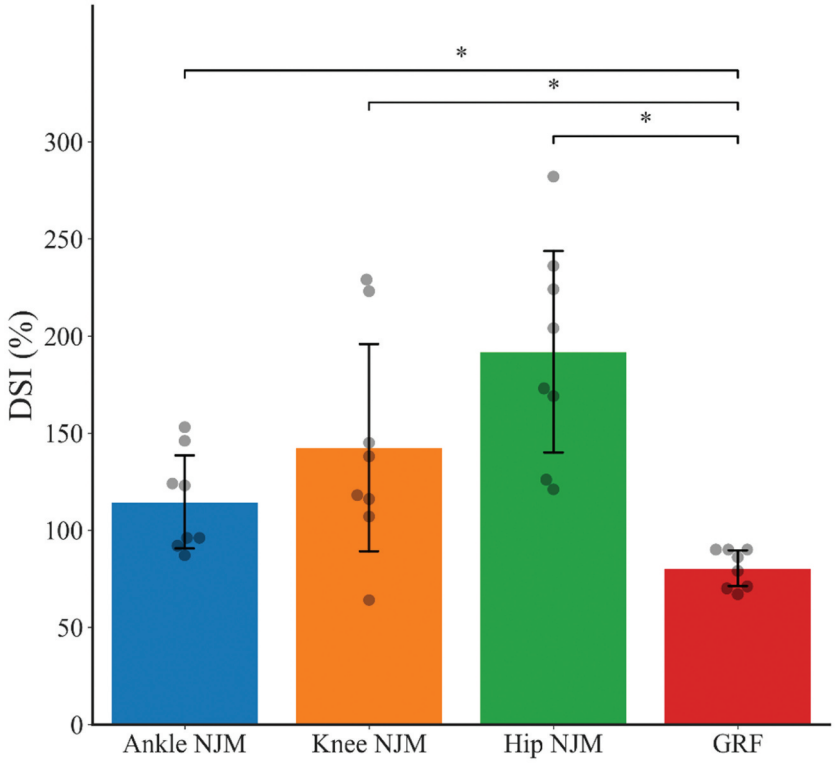

Figure 2. Mean±sd ground reaction force (GRF) and ankle, knee, hip net joint moment (NJM) dynamic strength indices (DSI [%]) based on the ratios of data from the isometric mid-thigh pull and counter-movement jumps. * indicates $p < 0.05$ vs. DSI_{GRF}.

Discussion and implications

The aims of this study were to determine the associations and differences between DSI_{GRF} and DSI_{NJM} values. We hypothesised that the traditional DSI_{GRF} values would correlate with at least one DSI_{NJM} and that these DSI values would not differ from each other. The results partially supported our initial hypothesis in that only the hip DSI_{NJM}

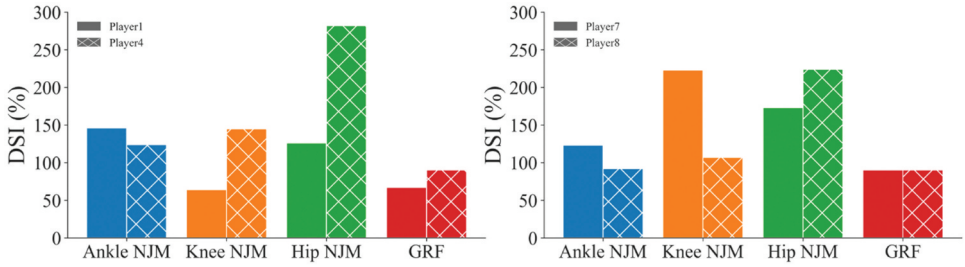

Figure 3. Examples of selected individual players with large or similar difference of ground reaction force (GRF) and relevant ankle, knee, hip net joint moment (NJM) dynamic strength indices (DSI [%]) Left: player with the highest (player 4: hatched bar) and lowest (player 1: sold bar) DSI_{GRF}. Right: players with similar DSI_{GRF} (player 7: solid bar, 8: hatched bar).

exhibited a significant positive correlation with DSI_{GRF}. In contrast, most athletes exhibited significantly greater DSI_{NJM} than DSI_{GRF} values, which conflicted with our initial hypothesis. While the current study is the first to calculate and report joint-based DSI values, these results suggest that DSI_{GRF} values are associated with hip DSI_{NJM}, the fact that joint-level DSI_{NJM} values were generally greater than DSI_{GRF} suggests that the ratio between peak NJM during the CMJ and IMTP may not provide intuitive or useful data for strength and conditioning coaches or sport scientists.

One of the main findings of this study was that that all DSI_{NJM} values were significantly greater than the DSI_{GRF} values, and appeared to increase in a distal-to-proximal direction. Specifically, the ankle, knee, and hip DSI_{NJM} were approximately 34%, 62%, and 111% greater than DSI_{GRF}. Since the average DSI_{NJM} for all joints exceeded 100% and the average DSI_{GRF} all fell below 100%, these findings indicate that the NJM that athletes were able to produce during the CMJ generally exceeded the NJM they were able to produce during the IMTP, whereas the reverse was true for the GRF. These findings may appear initially surprising because they indicate that the DSI_{GRF} reveals a deficit in an athlete's ability to effectively express a larger percentage of their overall maximum isometric strength during a dynamic task while the DSI_{NJM} values suggest the opposite. This discrepancy may reflect the fact that the joint angles at which athletes produce peak NJM during the CMJ differ from the joint angles during the IMTP. Specifically, hip and knee NJM during the CMJ often peak during the lowest position of the countermovement, with the hip and knee in significantly greater flexion than during the IMTP (Bobbert et al., 1986). In contrast, the joint angles during the IMTP are much less flexed, and while this position enables athletes to produce large GRF, it may not be ideal to enable athletes to generate large NJM (Ahn et al., 2021). These findings have important practical implications because DSI_{GRF} is often used by practitioners to inform training (Comfort et al., 2018, 2018; McMahon et al., 2017; Sheppard et al., 2011; Thomas et al., 2015). For example, seminal work by Sheppard et al. suggests that athletes who exhibit DSI_{GRF} greater than 0.8 should train to increase maximal strength whereas athletes with DSI_{GRF} less than 0.6 should focus on more effectively expressing their maximal strength during dynamics movements, potentially via speed-strength training (Sheppard et al., 2011). The suggestions made by Sheppard et al. were based on DSI_{GRF} of 0.7 that was calculated from the peak force between squat jump and isometric mid-thigh pull. While

the same authors also showed that DSI can track changes in effective force application during dynamic task with validity and high reliability, the authors explicitly acknowledged that the dynamics of the two tasks differed. Similarly, given that the mechanical demands imposed on the specific joints during dynamic tasks like the CMJ or isometric tasks like the IMTP may not be equivalent, greater joint-specific dynamic than isometric strength capacities could explain larger difference between DSI_{NJM} and DSI_{GRF}. The combination of these factors may thus make it possible for the DSI_{NJM} to exceed the DSI_{GRF} values, which may unfortunately suggest that DSI_{NJM} values might not be easily contextualised in the same sense as DSI_{GRF} and may not be practical for practitioners to evaluate an athlete's strength and monitor certain muscle adaptations to training.

Another finding of the current study was the hip DSI_{NJM} exhibited a very large positive correlation with the DSI_{GRF}. The presence and direction of this correlation may suggest that an athlete's ability to utilise a large percentage of their maximal isometric GRF generating capacity during a dynamic task is associated with their ability to utilise a greater percentage of their maximal isometric hip NJM generating capacity. It is important to note, however, that the hip DSI_{NJM} exceeded 100%, whereas DSI_{GRF} values were below 100%. Since no other studies have reported DSI_{NJM} data, it is difficult to reconcile the current findings with others in the literature. Despite the very large positive correlation between hip DSI_{NJM} and DSI_{GRF} the extent to which DSI_{GRF} could be used as a proxy of hip extensor muscle function, etc., is therefore debatable. In addition, neither ankle nor knee DSI_{NJM} was associated with DSI_{GRF}, which further suggests that the DSI_{GRF} provides limited information about joint-specific strength capacity. Therefore, while the theoretical concept of augmenting strength diagnostic assessments with predictions of joint-level DSI based on DSI_{GRF} data seems initially attractive, the results of the current study question its usefulness in practice.

The results from this study should be interpreted in light of several limitations. First, the conclusions of the current study are based on a small sample size, which may inadvertently affect the results, and it is therefore recommended that coaches and researchers interpret the current findings with caution. Second, the sample of participants was limited to one sex (females) and one sport (Lacrosse). Given the positioning and joint-specific function during the IMTP and CMJ, results may differ and not generalise to other populations of athletes based on, e.g., sex-based anthropometric differences, sport participation opportunities, or sport-specific training (Williams et al., 2018, 2018). Third, peak NJM during the IMTP was only calculated at one position, which may not reflect the global peak NJM that the ankle plantarflexor, knee, and hip extensor muscles are able to produce. In addition, the CMJ involves large ranges of joint motions and varying countermovement depths, which may affect the moment generating capacity of the muscles at each of the respective joints and not match the positioning of the body during the IMTP. Furthermore, the CMJ is a dynamic task that includes a stretch and shortening cycle whereas the IMTP does not. Collectively, all these elements may explain why the DSI_{NJM} sometimes exceeded 100%. Lastly, NJM only reflect the net effect of all flexor and extensor muscles as they produce forces across specific joints, which means that the function of individual agonist and antagonist muscle forces, and the effects of co-contraction, are ignored. In particular, neglecting the effects of muscle co-contraction likely underestimates the magnitudes of muscular contributions from the main extensor muscle groups to the NJM during

the IMTP and CMJ. Future research to address some of these limitations may include athletes from diverse sports, consider countermovement depth as a co-variate, use tasks other than the IMTP to assess maximum isometric performance (e.g., an isometric squat) and using more sophisticated musculoskeletal models that calculate muscle forces and account for co-contraction.

Conclusion

The current study is the first to calculate and report joint-based DSI values. The results showed that while the hip DSI_{NJM} was correlated with DSI_{GRF}, athletes generally exhibited greater DSI_{NJM} than DSI_{GRF} values. Although based on a small sample, these results suggest that while DSI_{GRF} values are associated with hip DSI_{NJM}, the fact that joint-level DSI_{NJM} values were generally greater than DSI_{GRF} suggests that the ratio between peak NJM during the CMJ and IMTP may not provide intuitive or useful data for strength and conditioning coaches or sport scientists.

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